In this paper, we address the above challenge and present a highly computationally efficient methodology for numerically simulating geoelectrical experiments in heterogeneous and complex fractured domains. Our approach builds on the recently developed 2-D discrete-dual-porosity (DDP) model for electric current flow in fractured media developed by Roubinet & Irving (2014), whereby fractures are explicitly represented using a semi-analytical formulation that takes into account the exchange of electric current flow between the discrete-fracture-network (DFN) and surrounding matrix. However, we importantly redevelop this formulation for the 2.5-D case, commonly considered in geoelectrical imaging, in order to accurately simulate current flow between point electrodes. The mathematical formulation of our new numerical method, including the general problem formulation in 2.5-D and the corresponding DDP modelling approach, is presented in Section 2. In Section 3, we validate our approach for both unfractured and fractured porous domains considering, in the latter case, both simple and complex fracture networks. Finally, we use our model in Section 4 to simulate electrical resistivity tomography (ERT) experiments in fractured porous domains composed of idealized and realistic fracture networks.

2 MODELLING APPROACH

2.1 General problem formulation

Consider a 3-D domain having electrical conductivity $\sigma(x, y, z)$ [S m^{-1}] in which an electric current I [A] is injected at position (x_0, y_0, z_0). Under steady-state conditions, the current flow in this domain is governed by the following charge-conservation equation at the point scale:

$$-\nabla \cdot [\sigma(x, y, z)\vec{\nabla}\phi(x, y, z)] = I\delta(x - x_0)\delta(y - y_0)\delta(z - z_0), \quad (1)$$

where $\phi(x, y, z)$ [V] is the electric potential and $\delta(.)$ [m^{-1}] is the Dirac delta function. Assuming that (i) the electrical conductivity σ is constant in the y-direction (i.e. $\sigma(x, y, z) = \sigma(x, z)$ and $\partial_y\sigma = 0$); (ii) the considered problem is symmetric in the y-direction (i.e. $\phi(x, y, z) = \phi(x, -y, z)$); and (iii) the current injection lies in the $y = 0$ plane (i.e. $y_0 = 0$), eq. (1) can be expressed in the Fourier domain as follows (e.g. Dey & Morrison 1979)

$$-\nabla \cdot [\sigma(x, z)\vec{\nabla}\bar{\phi}(x, \omega, z)] + \omega^2\sigma(x, z)\bar{\phi}(x, \omega, z) = \frac{I}{2}\delta(x - x_0)\delta(z - z_0), \quad (2)$$

where $\bar{\phi}(x, \omega, z)$ is the Fourier-cosine transform of $\phi(x, y, z)$ and ω is the wavenumber corresponding to the y-coordinate. The distributions of potential ϕ and $\bar{\phi}$ are related through (e.g. Bateman 1954):

$$\bar{\phi}(x, \omega, z) = \int_0^\infty \phi(x, y, z)\cos(\omega y)\mathrm{d}y \quad (3a)$$

$$\phi(x, y, z) = \frac{2}{\pi}\int_0^\infty \bar{\phi}(x, \omega, z)\cos(\omega y)\mathrm{d}\omega. \quad (3b)$$

Eq. (2) corresponds to the 2.5-D formulation of eq. (1), whereby the 3-D problem is decomposed into series of 2-D problems in the Fourier domain. That is, under the assumptions stated above, the 3-D electric potential $\phi(x, y, z)$ in eq. (1) can be determined by solving eq. (2) in the Fourier domain for several values of ω, and then inverting the resulting $\bar{\phi}(x, \omega, z)$ using the inverse Fourier-cosine transform (3b). Appendix A describes how this inverse Fourier-cosine transform is implemented and how the choice of wavenumber values is optimized in our work. The DDP formulation used to solve eq. (2) for heterogeneous and complex fractured domains is described next.

2.2 DDP approach

To develop a DDP formulation of the electric current flow problem (2) in the Fourier domain, we build upon the 2-D formulation presented by Roubinet & Irving (2014). In this formulation, the fractures and matrix are treated separately and coupled through the exchange of electric current between them. The fractures and matrix are discretized into fracture segments and matrix blocks having constant properties, respectively, and a linear system is created where the unknowns are the electrical potentials at the fracture intersections and extremities, as well as in the matrix blocks. Below, we derive the corresponding 2.5-D equations at the fracture-segment (Section 2.2.1), fracture-network (Section 2.2.2) and matrix-block (Section 2.2.3) scales. In doing this, it is assumed that fractures extend infinitely perpendicular to the 2-D modelling plane being considered. Note that our presentation contains only the key differences between this 2.5-D DDP formulation and the work of Roubinet & Irving (2014). For full information on the representation and discretization methods used to model the geological structures as well as on the solution of the linear system, please see their paper.

2.2.1 Electric potential along a fracture segment

For each 1-D fracture-segment k having constant aperture b_f^k and electrical conductivity σ_f^k, consider the charge conservation eq. (2) in the Fourier domain

$$-\sigma_f^k\partial^2_{x_f^k}\bar{\phi}_f^k + \omega^2\sigma_f^k\bar{\phi}_f^k = -\bar{Q}_{fm}^k, \quad (4)$$

where $\bar{\phi}_f^k = \bar{\phi}_f^k(x_f^k)$ is the Fourier-cosine transform of the electric potential averaged over the fracture aperture, x_f^k denotes the spatial variable along the fracture segment, and $\bar{Q}_{fm}^k$ is the Fourier-cosine transform of the source term related to the exchange of electric current between the fracture segment and the surrounding matrix. Considering that this fracture segment is located within matrix block (I_k, J_k), where $\bar{\phi}_m^{I_k,J_k}$ is the Fourier-cosine transform of the electric potential in this block, $\bar{Q}_{fm}^k$ can be expressed as

$$\bar{Q}_{fm}^k = -\alpha_{fm}^{I_k,J_k}\left(\bar{\phi}_m^{I_k,J_k} - \bar{\phi}_f^k\right). \quad (5)$$

Here, $\alpha_{fm}^{I_k,J_k}$ represents the fracture-matrix exchange coefficient, defined as $\alpha_{fm}^{I_k,J_k} = \sigma_m^{I_k,J_k}/d_{fm}^{I_k,J_k}$, where $\sigma_m^{I_k,J_k}$ is the matrix electrical conductivity of block (I_k, J_k) and $d_{fm}^{I_k,J_k}$ is the average normal distance between the fractures in that block and each point in the block (Roubinet & Irving 2014; Roubinet *et al.* 2016).

We consider Fourier-domain Dirichlet boundary conditions $\bar{\varphi}_f^{i_k}$ and $\bar{\varphi}_f^{j_k}$ at the extremities of each fracture segment $x_k = 0$ and $x_k = L_k$, respectively. Solving analytically eq. (4) with these conditions leads to the following expression for $\bar{\phi}_f^k$:

$$\bar{\phi}_f^k(x_k, \omega) = \beta_w(x_k)\bar{\varphi}_f^{i_k} + \frac{\gamma_w(x_k)}{\gamma_w(L_k)}\bar{\varphi}_f^{j_k} + \frac{\Gamma_{I_k,J_k}^k}{\Gamma_{I_k,J_k}^k + \omega^2}\left[1 - \beta_w(x_k) - \frac{\gamma_w(x_k)}{\gamma_w(L_k)}\right]\bar{\phi}_m^{I_k,J_k} \quad (6)$$

with

$$\Gamma_{I_k,J_k}^k \equiv \alpha_{fm}^{I_k,J_k}\Big/\left(b_f^k\sigma_f^k\right) \quad (7a)$$

$$\beta_w(x_k)=\exp\left(x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right)-\frac{\gamma_w(x_k)}{\gamma_w(L_k)}\exp\left(L_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right) \tag{7b}$$

$$\gamma_w(x_k)=\exp\left(-x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right)-\exp\left(x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right). \tag{7c}$$

2.2.2 *Modified DFN approach for the fracture network*

At the fracture-network scale, charge conservation at each fracture-intersection node is enforced by integrating eq. (2) over the intersection. For simplification, we consider that every node i is shared by N_i fracture segments having the same aperture b^i_f and conductivity σ^i_f, and that the surface of this intersection can be approximated by $b^i_f \times b^i_f$. Applying Gauss's Divergence theorem leads to

$$b^i_f\omega^2\sigma^i_f\bar{\phi}^i_{f|x^k_f=0}-\sigma^i_f\sum_{k=1}^{N_i}\partial_{x^k_f}\bar{\phi}^k_{f|x^k_f=0}=0. \tag{8}$$

Using expression (6), eq. (8) can be rewritten as

$$b^i_f\omega^2\sigma^i_f\bar{\varphi}^{i_k}_f-\sigma^i_f\sum_{k=1}^{N_i}\left(A_{i_k}\bar{\varphi}^{i_k}_f+A_{j_k}\bar{\varphi}^{j_k}_f+A_{I_k,J_k}\bar{\phi}^{I_k,J_k}_m\right)=0, \tag{9}$$

where the terms A_{i_k}, A_{j_k} and A_{I_k,J_k} are defined as

$$A_{i_k}=\zeta_w(x_k)\sqrt{\Gamma^k_{I_k,J_k}+\omega^2} \tag{10a}$$

$$A_{j_k}=-\frac{\lambda_w(x_k)}{\gamma_w(L_k)}\sqrt{\Gamma^k_{I_k,J_k}+\omega^2} \tag{10b}$$

$$A_{I_k,J_k}=-\frac{\Gamma^k_{I_k,J_k}}{\Gamma^k_{I_k,J_k}+\omega^2}\left(A_{i_k}+A_{j_k}\right) \tag{10c}$$

with

$$\zeta_w(x_k)=\exp\left(x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right)+\frac{\lambda_w(x_k)}{\gamma_w(L_k)}\exp\left(L_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right) \tag{11a}$$

$$\lambda_w(x_k)=\exp\left(x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right)+\exp\left(-x_k\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}\right). \tag{11b}$$

2.2.3 *Modified finite-volume approach in the matrix*

Finally, in the matrix, charge conservation is enforced in the Fourier domain by integrating eq. (2) over each matrix block (I, J) of volume $V_{I,J}$. This leads to

$$-\int_{V_{I,J}}\nabla\cdot\left(\sigma_m\vec{\nabla}\bar{\phi}^{I,J}_m\right)\mathrm{d}V+\int_{V_{I,J}}\omega^2\sigma_m\bar{\phi}^{I,J}_m\mathrm{d}V=\int_{V_{I,J}}\bar{Q}^k_{fm}\mathrm{d}V. \tag{12}$$

Using Gauss' Divergence Theorem, the left-hand side of eq. (12), which we denote as $M_{I,J}$, can be discretized as

$$\begin{aligned}M_{I,J}&=C_{I,J}\bar{\phi}^{I,J}_m+C^W_{I,J}\bar{\phi}^{I-1,J}_m+C^E_{I,J}\bar{\phi}^{I+1,J}_m\\&\quad+C^S_{I,J}\bar{\phi}^{I,J-1}_m+C^N_{I,J}\bar{\phi}^{I,J+1}_m,\end{aligned} \tag{13}$$

where

$$C^W_{I,J}=-\frac{\Delta z}{\Delta x}\mathcal{H}_{(I-1,J),(I,J)} \tag{14a}$$

$$C^E_{I,J}=-\frac{\Delta z}{\Delta x}\mathcal{H}_{(I+1,J),(I,J)} \tag{14b}$$

$$C^S_{I,J}=-\frac{\Delta x}{\Delta z}\mathcal{H}_{(I,J-1),(I,J)} \tag{14c}$$

$$C^N_{I,J}=-\frac{\Delta x}{\Delta z}\mathcal{H}_{(I,J+1),(I,J)} \tag{14d}$$

$$C_{I,J}=\omega^2\sigma^{I,J}_m\Delta x\Delta z-C^W_{I,J}-C^E_{I,J}-C^S_{I,J}-C^N_{I,J} \tag{14e}$$

with $\mathcal{H}_{(K,L),(I,J)}$ the harmonic mean of the electrical conductivity in matrix blocks (K, L) and (I, J), i.e. $\mathcal{H}_{(K,L),(I,J)}=2/\left(1/\sigma^{K,L}_m+1/\sigma^{I,J}_m\right)$.

The right-hand side of eq. (12) can be expressed as

$$\int_{V_{I,J}}\bar{Q}^k_{fm}\mathrm{d}V=\sum_{k=1}^{N^f_{I,J}}\int_0^{L_k}\bar{Q}^k_{fm}\mathrm{d}V, \tag{15}$$

where $N^f_{I,J}$ is the number of fractures contained in the matrix block volume $V_{I,J}$. Using expression (5) for the source term $\bar{Q}^k_{fm}$ leads to

$$\int_{V_{I,J}}\bar{Q}^k_{fm}\mathrm{d}V=-\alpha^{I,J}_{fm}\bar{\phi}^{I,J}_m\sum_{k=1}^{N^f_{I,J}}L_k+\alpha^{I,J}_{fm}\sum_{k=1}^{N^f_{I,J}}\bar{\Phi}^k_f, \tag{16}$$

where $\bar{\Phi}^k_f$ is the integrated value of $\bar{\phi}^k_f$ along fracture segment k, i.e. $\bar{\Phi}^k_f=\int_0^{L_k}\bar{\phi}^k_f\mathrm{d}x_k$, and $(I_k, J_k) = (I, J)$ for $k = 1,\ldots,N^f_{I,J}$. Integrating expression (6) for $\bar{\phi}^k_f$, we obtain the following definition for $\bar{\Phi}^k_f$:

$$\bar{\Phi}^k_f=D_{i_k}\bar{\varphi}^{i_k}_f+D_{j_k}\bar{\varphi}^{j_k}_f+D_{I,J}\bar{\phi}^{I,J}_m, \tag{17}$$

where the coefficients D_{i_k}, D_{j_k}, and $D_{I,J}$ are defined as

$$D_{i_k}=\frac{\zeta_w(L_k)-1}{\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}}-\frac{2\exp\left(\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}L_k\right)}{\gamma_w(L_k)\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}} \tag{18a}$$

$$D_{j_k}=\frac{2-\lambda_w(L_k)}{\gamma_w(L_k)\sqrt{\Gamma^k_{I_k,J_k}+\omega^2}} \tag{18b}$$

$$D_{I,J}=\frac{\Gamma^k_{I,J}}{\Gamma^k_{I,J}+\omega^2}\left(L_k-D_{j_k}-D_{i_k}\right). \tag{18c}$$

Finally, the discretized expression of eq. (12) is given by

$$\begin{aligned}&\left[C_{I,J}+\alpha^{I,J}_{fm}\sum_{k=1}^{N^f_{I,J}}(L_k-D_{I,J})\right]\bar{\phi}^{I,J}_m+C^W_{I,J}\bar{\phi}^{I-1,J}_m+C^E_{I,J}\bar{\phi}^{I+1,J}_m\\&+C^S_{I,J}\bar{\phi}^{1,J-1}_m+C^N_{I,J}\bar{\phi}^{I,J+1}_m-\alpha^{I,J}_{fm}\sum_{k=1}^{N^f_{I,J}}\left(D_{i_k}\bar{\varphi}^{i_k}_f+D_{j_k}\bar{\varphi}^{j_k}_f\right)=0.\end{aligned} \tag{19}$$

3 VALIDATION

We now validate our 2.5-D modelling approach for unfractured (Section 3.1) and fractured (Section 3.2) porous domains considering a variety of different boundary conditions. We begin with simple configurations for which known analytical solutions exist (Sections 3.1 and 3.2.1). We then validate our approach for more complex configurations involving multiple fractures using a standard finite-element approach as a reference solution (Section 3.2.2).

3.1 Unfractured porous domains

Validating on unfractured porous domains enables us to verify the modified finite-volume formulation presented in Section 2.2.3. Here we consider the homogeneous and two-layer